



- ① What do you mean by Simple Ether and mixed ether? How is common ether prepared in the lab & in dry and pure form. Discuss the properties and mention the uses.
- ② Describe the preparation of Di-Ethyl Ether from Ethanol. Why is the method called "continuous Etherification process"? Sketch the diagram of apparatus ^{you would use} and explain the mechanism of the reaction. What variation in the experimental condition would you make to obtain an entirely different products. Distinguish between C_2H_5OH & $C_2H_5OC_2H_5$.

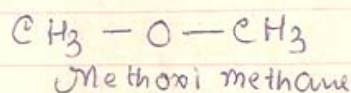
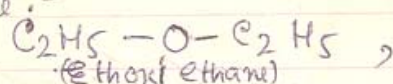
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When Oxygen atom is linked with two same or different alkyl groups we call it ether i.e. $R-OR$.

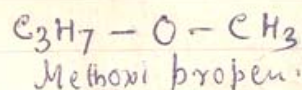
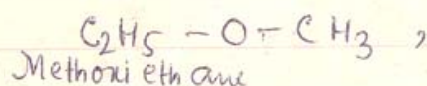
When R, the alkyl group be the same we call it simple ether, and when Radicals be the different we call it mixed ether.

Examples:-

Simple:-



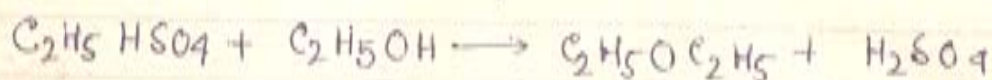
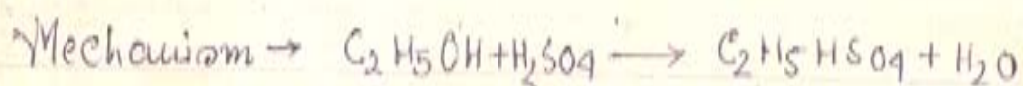
Mixed ether



However $C_2H_5-O-C_2H_5$, although called Diethyl ether it is also called simple ether as well as common ether and briefly called ether. It may be called ethoxy ethane.



Theory $\rightarrow$ When a mixture of Conc. H_2SO_4 and excess of C_2H_5OH mixture is heated at $140^\circ C$ we get ether.



Procedure $\rightarrow$ In a distilling flask about 100 cc of C_2H_5OH is taken. With the help of dropping funnel about 50 cc of Conc. H_2SO_4 is poured into it. The flask is fitted with a thermometer to record the temperature $140^\circ C$. It is fitted with a water condenser which is connected to a receiver via an adaptor. The receiver is kept immersed in ice bath. The receiver is connected to a glass tube then a rubber tube ending to a longer distance from the flame as shown in the diagram.

On heating the mixture nearly over sand bath at $140^\circ C$ (nearly) we get the vapour of ether, which is cooled by water condenser to get liquid ether. It is collected in the bottle. The uncondensed vapour is sent to a long distance to avoid explosion.

Precaution $\rightarrow$ Since ether is highly inflammable, a screen is put in between the flame and receiver.



Purification → This sample may contain SO_2 & CO_2 dissolved (H_2SO_3 and H_2CO_3) so the sample is treated with dilute KOH/NaOH solⁿ in a separating funnel to remove the above impurities. Then it is treated with saturated solution of NaCl in a separating funnel to remove alcohol.

The little quantity of water present is removed by putting a piece of solid fused CaCl_2 .

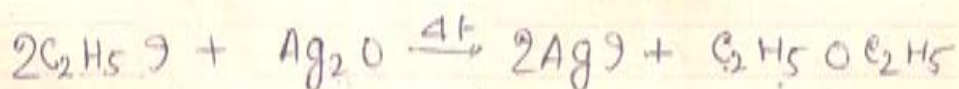
If 100% pure ether is required it is redistilled again at about 50°C in water bath.

Other Method Of Preparation

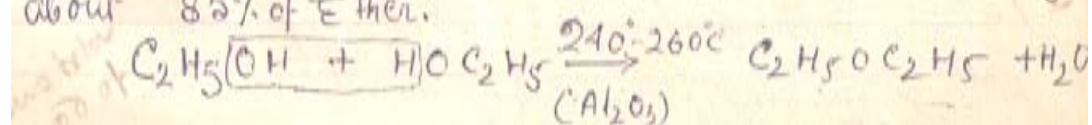
① When Ethyl halide/ethyl iodide is heated with Sodium ethoxide we get ether and this reaction is called Williamson Synthesis.

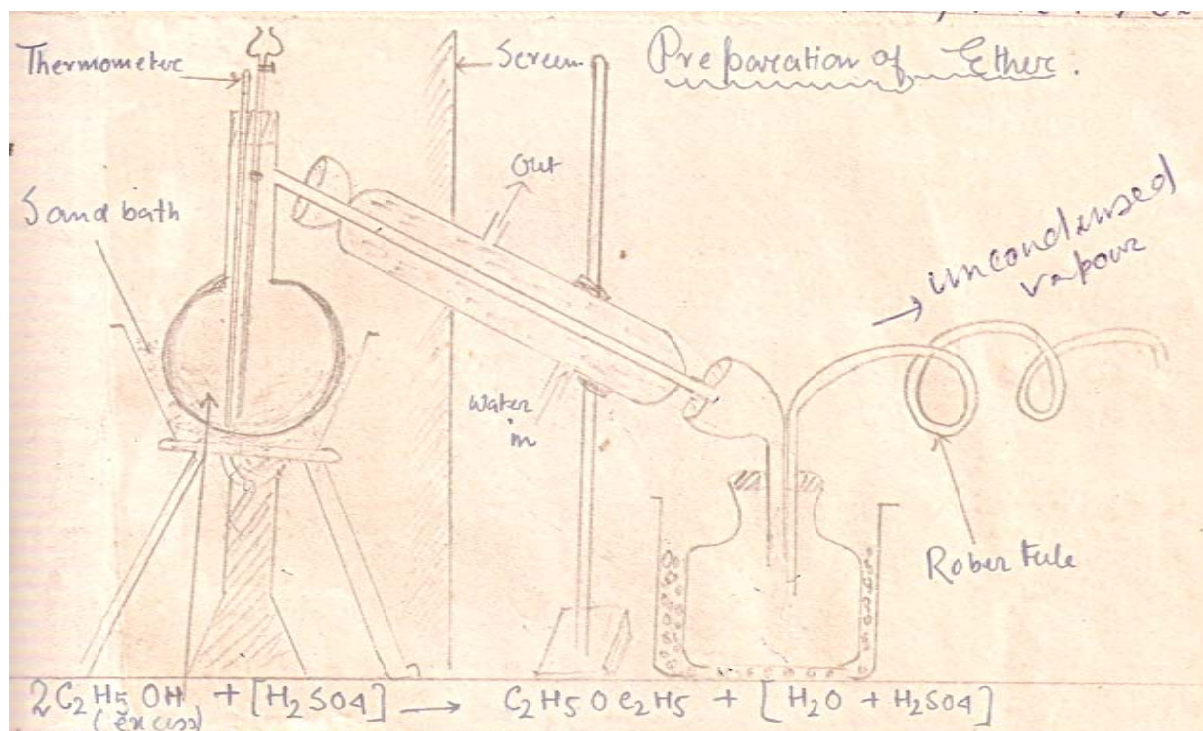


② W.H When Ethyl halide/ethyl iodide is heated with Silver oxide (Ag_2O) ether is formed.



③ When vapour of Ethyl alcohol is passed over heated Alumina (Al_2O_3) at $240^\circ - 260^\circ\text{C}$ we get about 85% of Ether.







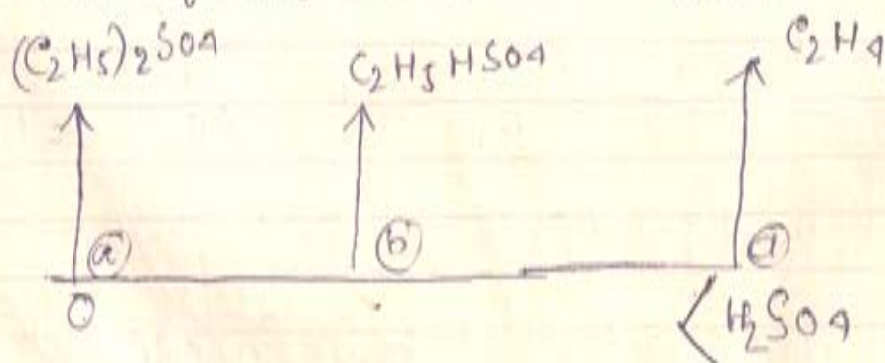
NP Note ON Continuous Etherification

Ether is prepared in the lab by heating a mixture of conc. H_2SO_4 and excess of C_2H_5OH at $140^\circ C$. After having ether, conc. H_2SO_4 is left in the distilling flask now fresh C_2H_5OH is added through dropping funnel into the distilling flask and repeating the process we get ether again, like that we can get ether again and again and this is known as Continuous Etherification.

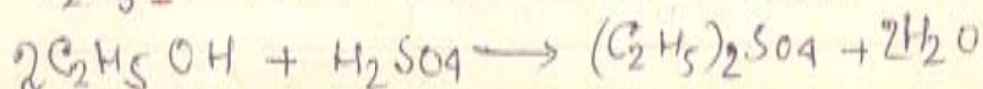
However in the lab practically we may prepare 6 or 7 times because H_2SO_4 is diluted gradually due to the formation of water molecules in the reaction and also some of sulphuric acid is converted into SO_2 and SO_3 etc.

What products on variation.

Reaction of C_2H_5OH & conc H_2SO_4

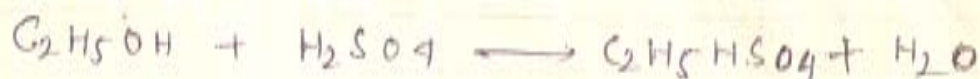


(a) At low temperature we get Diethyl Sulphate when C_2H_5OH is mixed with conc. H_2SO_4

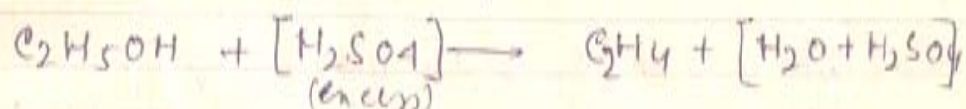




- ⑤ On heating the above mixture at 100°C we get Ethyl hydrogen Sulphate.



- ⑥ Heating the above mixture, conc H_2SO_4 in excess at 165°C we get Ethylene

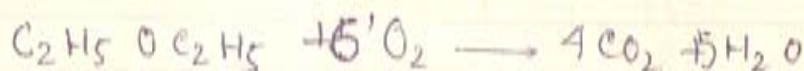


Physical Properties

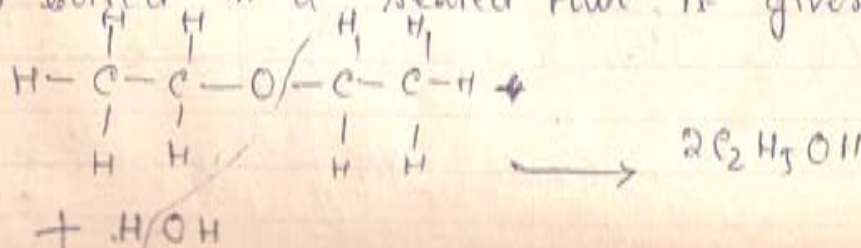
It is a volatile liquid. Sp. gravity less than 1 (0.72) insoluble in water so it floats in water, pleasant smell, sweetish burning taste. It is soluble in alcohol and it is a very good solvent for many organic substances.

CHEM. PROP

- (1) Burning or Oxidation $\rightarrow$ It burns with a clear green flame producing CO_2 and H_2O

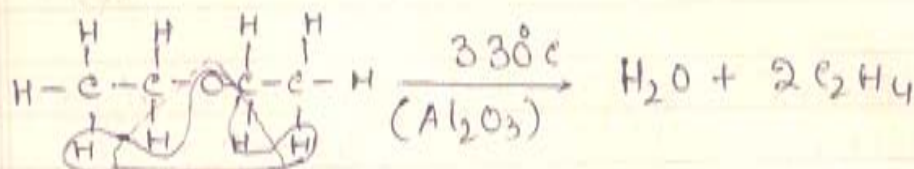


- (2) Hydrolysis $\rightarrow$ When ether is mixed with water and boiled in a sealed tube it gives alcohol.

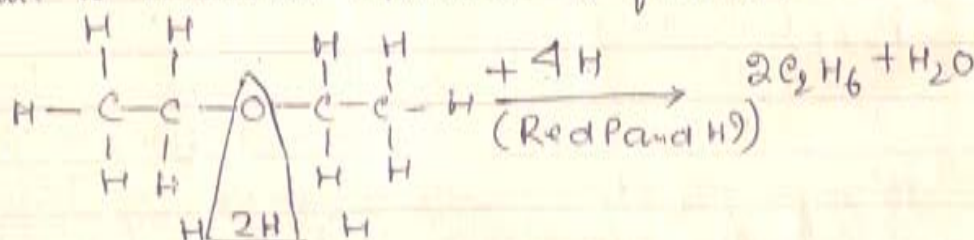




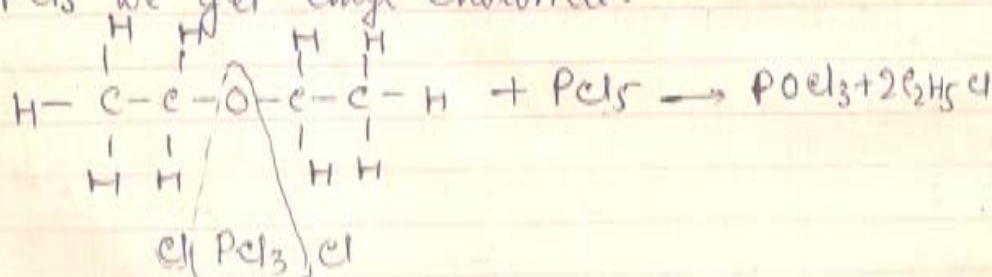
(3) With Alumina $\rightarrow$ When Vapour of Ether is passed over heated Alumina (Al_2O_3), as a catalyst it gives ethylene at 330°C



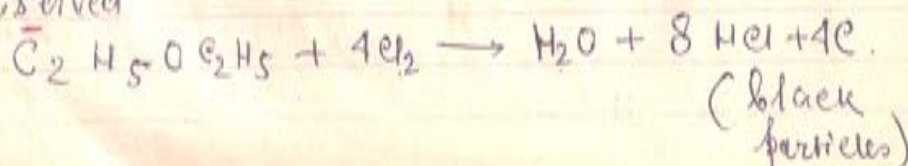
(4) With red P and H_2 / Reducing agent $\rightarrow$ When ether is treated with red phosphorus and H_2 due to reduction ethane is formed.



(5) With Pcl_5 $\rightarrow$ When ether is heated with Pcl_5 we get ethyl chloride.

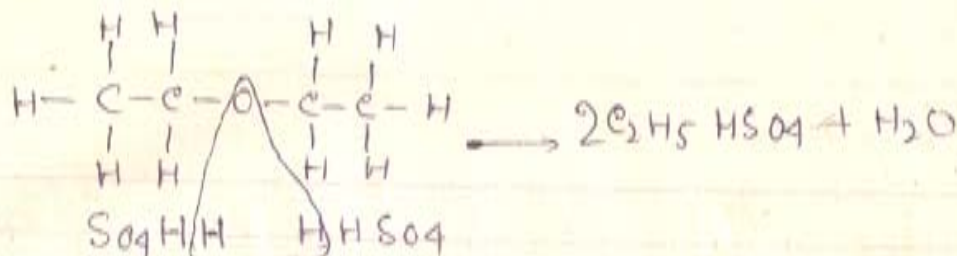
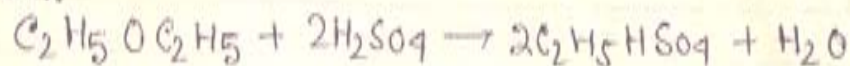


(6) With Cl_2 gas $\rightarrow$ When a piece of blotting paper moistened with ether is introduced into a gas jar of Cl_2 , it at once catches fire (without match) and small black particles are observed





- ⑦ With Cone. $H_2SO_4 \rightarrow$ When ether is heated with Cone. H_2SO_4 Ethyl hydrogen Sulphate is formed.



Uses of Ether

- (1) It is a good solvent for oils, fat, gum and Resin.
- ✓ (2) In hospital as an anesthetic in surgery.
- (3) As a refrigerant (cooling agent) (A mixture of ether and dry ice (CO_2 solid) gives a low temperature as $-77^\circ C$)
- (4) As a fuel in place of Petrol after mixing it with alcohol.
- * (5) "Ether extract" in industry. for extracting organic compounds from their natural source.

Distinction between C_2H_5OH & $C_2H_5OC_2H_5$	
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Reagent/Property	C_2H_5OH (Ethyl alcohol)
1. Smell	Alcohol. Smell
2. Solubility in water	Completely soluble in water in any ratio
3. With Na	Gives H_2 and C_2H_5ONa
4. With PCl_5	$POCl_3$ & HEl are obtained
5. With I_2 and $NaOH$	Yellow Crystals of CHI_3
	$C_2H_5OC_2H_5$ (Ether)
	Pleasant
	Insoluble.
	No action
	Only $POCl_3$
	No such